

Nandan Bharatkumar Parikh

(404)-883-8113 | nandanbparikh@gmail.com | linkedin.com/in/nandanbparikh | github.com/lelouch0204

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science (ML Specialization) | **GPA: 4.0/4.0**

August 2025 – May 2027

- Relevant Coursework: Computer Vision, Machine Learning, Efficient ML

Birla Institute of Technology and Science, Pilani

Pilani, India

Bachelor of Engineering in Computer Science

August 2019 – May 2023

- GPA: 8.8/10, Major GPA: 9.48/10
- Relevant Coursework: Deep Learning, Applied Statistical Methods, Image Processing, Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms, Compiler Construction

RESEARCH EXPERIENCE

Graduate Research Assistant

January 2026 – Present

Georgia Tech (Prof. James Hays) | Video Generation, 3D Vision

Atlanta, GA

- Physically Realistic Video Generation:** Architecting a pipeline to convert static images into dynamic 3D scenes using 3D Gaussian Splatting and SuGaR for high-fidelity mesh extraction.
- Simulating rigid-body dynamics in NVIDIA Isaac Sim to enforce physical constraints (collision, gravity) on Generative AI (Diffusion) video outputs, ensuring temporal consistency.

Graduate Student Researcher

August 2025 – Present

Georgia Institute of Technology (Bio-MIBLab, Advisor: Prof. May Wang)

Atlanta, GA

- Investigating hierarchical editing mechanisms for concept bottleneck models to improve explainability in vision systems
- Developing methods to enable localized concept editing where only semantically related concepts are affected
- Conducting experiments on CUB and CelebA-HQ datasets for fine-grained visual recognition tasks

Research Intern | [Publication]

November 2022 – June 2023

Harvard University | Computer Vision, High-Performance Computing

Boston, MA

- Real-Time Volumetric Restoration:** Engineered a **Graph Reasoning Attention Network (GNN)** pipeline to restore structural fidelity in large-scale **3D Volumetric** data streams.
- Custom CUDA Optimization:** Achieved **20x inference speedup** by implementing custom **CUDA kernels** for latency-critical processing, enabling high-throughput handling of massive sensor datasets.
- Developed automated data acquisition pipelines that filter low-quality frames in real-time, reducing downstream storage and compute costs.

INDUSTRY EXPERIENCE

Software Engineer II (SDE-2)

July 2023 – July 2025

Flipkart (Walmart) | Hotels Search & Recommendation

Bengaluru, India

- Open Search:** Architected a **RAG-based search system** serving 5k+ users using **Gemini** and **LangChain4j**, integrating hybrid vector retrieval to maximize result relevance.
- Vector Search Infrastructure:** Evaluated vector databases (Qdrant, Pinecone) and optimized existing Elasticsearch cluster to avoid \$20K annual infra costs while meeting search performance requirements
- Scaled microservices to **24k RPS (5x growth)** via JVM tuning and offline **cache warming** (Redis and Aerospike), ensuring high availability during peak traffic events.
- Engineered **Semantic Caching** using vector similarity and keyword matching to capture fuzzy queries, improving **cache hit rate by 30%** and reducing redundant LLM costs.
- Built **BigQuery ETL pipelines** to compute **CTR and popularity metrics**, indexing data into Elasticsearch to optimize autosuggest and reduce zero-result queries by **35%**.
- Deployed **Prompt Engineering pipelines** for intent parsing and metadata extraction, significantly improving accuracy on unstructured natural language queries.

Software Engineering Intern

June 2022 – August 2022

Flipkart

Bengaluru, India

- Built Python security automation framework integrating MobSF across internal mobile app pipeline
- Developed self-service web portal with Google Apps Script reducing manual security audits by 60%

TEACHING EXPERIENCE

Graduate Teaching Assistant – Compilers and Interpreters

Jan 2026 – May 2026

Georgia Institute of Technology

Atlanta, GA

- Created and graded evaluative components relating to DFAs, Context-Free Languages and Parsers
- Conducted weekly office hours and pre-exam recitals to resolve student doubts

Teaching Assistant – Neural Networks and Fuzzy Logic

August 2022 – December 2022

BITS Pilani

Pilani, India

- Curated and shortlisted research papers for student term projects in neural networks and fuzzy systems
- Mentored students in preparing research-based term papers and implementation projects

PROJECTS

Distributed Training Implementation (FSDP & DDP) | *PyTorch, NVIDIA H100s, CUDA*

GitHub

- Implemented **Fully Sharded Data Parallel (FSDP)** and Pipeline Parallelism on **NVIDIA H100** clusters to scale transformer training.
- Achieved **96.8% scaling efficiency** (near-linear) across multi-GPU nodes by optimizing pipeline micro-batching and masking communication overhead.
- Benchmarked FSDP vs. DDP, maintaining **99.8% baseline throughput** while reducing memory fragmentation.

Efficient LLM Inference & Quantization | *PyTorch, Llama-3, SmoothQuant*

GitHub

- Enabled **W8A8 inference** on Llama-3 via SmoothQuant, preserving FP16 fidelity (6.24 PPL) despite 40x activation outliers
- Analyzed Attention Sinks, finding initial tokens capture 95% probability mass in deep layers to validate StreamingLLM mechanics
- Profiled KV Cache dynamics, quantifying linear memory growth at 0.125 MB/token for long-context optimization

Water Quality Time Series Analysis | *Python, Pandas, Scikit-learn*

GitHub

- Built end-to-end pipeline for forecasting pollutants in water using time series decomposition and ADF stationarity testing
- Compared VARIMAX, Linear Regression, Random Forests, LSTMs, and Transformers, achieving best MAPE of 7.90%

TECHNICAL SKILLS

Languages & Data: Python, Java, C++, SQL, C, JavaScript, Redis, MongoDB, Elasticsearch

Machine Learning: PyTorch, TensorFlow, Transformers, LangChain, OpenCV, Scikit-learn, Pandas

Backend & Cloud: Spring Boot, Docker, Kubernetes, GCP, BigQuery, Kafka, Microservices, Git

PUBLICATIONS & PRESENTATIONS

Real-time Image Deblurring to Improve Throughput of Serial-Section Volume EM

2023

R. L. Schalek, N. Parikh, Y. Wu, J. W. Lichtman, and D. Wei

- Microscopy and Microanalysis, vol. 29, no. S1, pp. 988-989 (Poster Presentation)

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Association of Computing Machinery (ACM), BITS Pilani

Sophomore – Junior Year

Core Member, Game Development Lead

- Developed program connecting students with experts in niche technical fields, bridging gaps in research
- Organized alumni-led talks on diverse career paths, providing students with mentorship and insights
- Led creation of game showcased at university's technical fest APOGEE, enhancing visibility and engagement
- Recruited and mentored students in game development, building a sustainable technical community

AWARDS & HONORS

Qualified for Indian National Mathematical Olympiad (INMO), 2018

Ranked 6th out of 80 teams in Compiler Construction Competition, BITS Pilani, 2022

Promoted early from SDE-I to SDE-II at Flipkart for exceptional backend contributions, 2024